

Q-1 $\lim_{x \rightarrow 0} \frac{e - (1+x)^{1/x}}{\tan(x)}$

Sol $k = e^{\ln(k)}$
 $a^b = e^{\ln(a^b)} = e^{b \ln(a)}$

$\lim_{x \rightarrow 0} \frac{e - e^{\frac{\ln(1+x)}{x}}}{\tan(x) \cdot x}$

$= \lim_{x \rightarrow 0} -e \left(e^{\frac{\ln(1+x)}{x} - 1} - 1 \right)$

$= \lim_{x \rightarrow 0} -e \left(\frac{e^{\frac{\ln(1+x)}{x} - 1} - 1}{\frac{x}{2}} \right) = \lim_{x \rightarrow 0} -e \left(\frac{\ln(1+x) - x}{x^2} \right)$

$= \lim_{x \rightarrow 0} \frac{e - x}{2 \cdot x^2} = \boxed{\frac{e}{2}}$

Q-2 Find $a, b, c, d \in \mathbb{R}$ if

$\lim_{x \rightarrow \infty} \left(\sqrt{x^4 + ax^3 + 3x^2 + bx + 2} - \sqrt{x^4 + 2x^3 - cx^2 + 3x - d} \right) = \underline{4}$

$= \lim_{x \rightarrow \infty} \frac{(a-2)x^3 + (3+c)x^2 + (b-3)x + 2+d}{\sqrt{x^4 + ax^3 + 3x^2 + bx + 2} + \sqrt{x^4 + 2x^3 - cx^2 + 3x - d}}$

$= \lim_{x \rightarrow \infty} \frac{(a-2)x^3 + (c+3)x^2 + (b-3)x + 2+d}{2x^2}$

$a = 2$

$\frac{c+3}{2} = 4 \quad \text{①}$
 $a = 2 \quad \text{②} \quad c = 5 \quad \text{③} \quad b \in \mathbb{R} \quad \text{④} \quad d \in \mathbb{R}$

Short cut

$$x \rightarrow 0.$$

$$\sin(x), \tan(x), \tan^{-1}(x), \sin^{-1}(x), \\ \ln(1+x), e^x - 1 \longrightarrow 1.x$$

$$\frac{\ln(1+x)}{x} = 1$$

$$\frac{e^x - 1}{x} = 1$$

$$1 - \cos(A) \xrightarrow{0} \frac{A^2}{2}$$

$$\frac{1 - \cos(x)}{x^2} = \left(\frac{1}{2}\right)$$

~~$\frac{\sin(x) - x}{x^3}$~~

Que $\lim_{x \rightarrow 0^+} \frac{\log_{\sin(x)}(\cos(x))}{\log_{\sin(\frac{x}{2})}(\cos(\frac{x}{2}))}$

Sol

$$\lim_{x \rightarrow 0^+} \frac{\left(\frac{\ln(\cos(x))}{\ln(\sin(x))} \right)}{\left(\frac{\ln(\cos(\frac{x}{2}))}{\ln(\sin(\frac{x}{2}))} \right)}$$

$$= \lim_{x \rightarrow 0^+} \frac{\overset{0}{\ln(\cos(x))} \cdot \overset{-\infty}{\ln(\sin(\frac{x}{2}))}}{\underset{-\infty}{\ln(\sin(x))} \cdot \underset{0}{\ln(\cos(\frac{x}{2}))}}$$

$$= \lim_{x \rightarrow 0} \left(\frac{\ln(\cos(x))}{\ln(\cos(\frac{x}{2}))} \right) \cdot \left(\frac{\ln(\sin(\frac{x}{2}))}{\ln(\sin(x))} \right)$$

Que $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos^2(x)}}{\sqrt{x^2 + 8x^3}}$

$$= \lim_{x \rightarrow 0} \frac{\sqrt{\sin^2(x)}}{\sqrt{x^2} \sqrt{1+8x}} = 1$$

$$= \lim_{x \rightarrow 0} \frac{|\sin(x)|}{|x|}$$

$$= \lim_{x \rightarrow 0} \left| \frac{\sin(x)}{x} \right| = 1$$

Que $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos^2(x)}}{x}$

A $\lim_{x \rightarrow 0} \frac{(\sin(x))}{x} = D.N.E$

Que $\lim_{x \rightarrow 0} g^{\frac{1}{\cos(x)-1}}$

$$= \lim_{x \rightarrow 0} (g)^{\frac{\cos(x)-1}{\cos(x)-1}} = \boxed{0}$$

$x \rightarrow 0 \Rightarrow \cos(x) \rightarrow 1^-$
 $\cos(x)-1 \rightarrow 0^-$
 $\frac{1}{\cos(x)-1} \rightarrow -\infty$
 $(g)^{\frac{1}{\cos(x)-1}} \rightarrow 0$

$$\lim_{x \rightarrow 0^+} \left(\frac{\ln(\cos x)}{\ln(\cos(\frac{x}{2}))} \right) \left(\frac{\ln(\sin(\frac{x}{2}))}{\ln(\sin(x))} \right) \rightarrow 2 \sin(\frac{x}{2}) \cos(\frac{x}{2})$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{\cos(x) - 1}{\cos(\frac{x}{2}) - 1} \right) \cdot \left(\frac{\ln(\sin(\frac{x}{2}))}{\ln(\sin(\frac{x}{2})) + \ln(2\cos(\frac{x}{2}))} \right)$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{2 \sin^2(\frac{x}{2})}{2 \sin^2(\frac{x}{4})} \right) \cdot \left(\frac{1}{1 + \frac{\ln(2\cos(\frac{x}{2}))}{\ln(\sin(\frac{x}{2}))}} \right)$$

$$\ln(1+x) \xrightarrow{x \rightarrow 0} x - 1$$

$$\ln(A) \rightarrow A - 1$$

$\rightarrow 1$

$$= \frac{\left(\frac{x}{2}\right)^2}{\left(\frac{x}{4}\right)^2} \cdot 1$$

$$= \boxed{4} A$$

Que Let $S_n = \sin(\theta) + \sin(2\theta) + \sin(3\theta) + \dots + \sin(n\theta)$, given $\sin(\frac{\theta}{2}) \neq 0$.
 Evaluate $\lim_{n \rightarrow \infty} \frac{S_1 + S_2 + S_3 + \dots + S_n}{n}$

Soln $S_n = \sin(\theta) + \sin(2\theta) + \dots + \sin(n\theta)$

$$= \frac{\sin\left(n \cdot \frac{\theta}{2}\right) \cdot \sin\left(\frac{\theta + n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)} = \frac{\cos\left(\frac{\theta}{2}\right) - \cos\left(\frac{\theta}{2} + n\theta\right)}{2 \sin\left(\frac{\theta}{2}\right)} = \frac{1}{2} \cot\left(\frac{\theta}{2}\right) - \frac{1}{2 \sin\left(\frac{\theta}{2}\right)} \cdot \cos\left(\frac{\theta}{2} + n\theta\right)$$

$$S_1 + S_2 + \dots + S_n = \sum_{r=1}^n S_r = \sum_{r=1}^n \left(\frac{1}{2} \cot\left(\frac{\theta}{2}\right) - \frac{\cos\left(\frac{\theta}{2} + r\theta\right)}{2 \sin\left(\frac{\theta}{2}\right)} \right)$$

$$= \frac{n}{2} \cot\left(\frac{\theta}{2}\right) - \frac{1}{2 \sin\left(\frac{\theta}{2}\right)} \left(\frac{\sin\left(n \cdot \frac{\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)} \cdot \cos\left(\frac{\frac{\theta}{2} + \theta + \frac{\theta}{2} + n\theta}{2}\right) \right)$$

$$\lim_{n \rightarrow \infty} \frac{s_1 + s_2 + \dots + s_n}{n} = \lim_{n \rightarrow \infty} \frac{\cancel{\frac{1}{2}} \cot\left(\frac{\theta}{2}\right)}{\cancel{\frac{1}{2}}} = \frac{1}{2} \cot\left(\frac{\theta}{2}\right)$$

Q.E.D.

$$= \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{n\theta}{2}\right) \cdot \cos\left(\theta + \frac{n\theta}{2}\right)}{2 \sin^2\left(\frac{\theta}{2}\right) \cdot n}$$

Q.E.D.

Ques $\lim_{n \rightarrow \infty} \underbrace{\left(\underbrace{3^n}_{\infty} + \underbrace{5^n}_{\infty} + \underbrace{7^n}_{\infty} \right)^{\frac{1}{n}}}_{\infty}$

$= \lim_{n \rightarrow \infty} 7 \left(\left(\frac{3}{7} \right)^n + \left(\frac{5}{7} \right)^n + 1 \right)^{\frac{1}{n}}$

$= \boxed{7}$ Ans

Rem

$$\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln(a)$$

Ques $\lim_{n \rightarrow \infty} n^2 \left(\sqrt[n]{a} - \sqrt[n+1]{a} \right)$,
given $n \in \mathbb{N}$ & $a \in \mathbb{R}^+$

$= \lim_{n \rightarrow \infty} n^2 \left(a^{\frac{1}{n}} - a^{\frac{1}{n+1}} \right)$

$= \lim_{n \rightarrow \infty} n^2 \cdot \left(a^{\frac{1}{n} - \frac{1}{n+1}} - 1 \right)$

$= \lim_{n \rightarrow \infty} \frac{n^2}{\frac{1}{n(n+1)}} \left(\frac{a^{\frac{1}{n(n+1)}} - 1}{\left(\frac{1}{n(n+1)} \right)} \right)$

$= \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}} \cdot \ln(a)$

$= \underline{\underline{\ln(a)}}$ Ans

$\ln(a)$

Ques $\lim_{x \rightarrow 1} \left(\frac{p}{1-x^p} - \frac{q}{1-x^q} \right), p, q \in \mathbb{N}$

$$= \lim_{x \rightarrow 1} \left(\frac{p}{1-x^p} - \frac{q}{1-x^q} \right) \neq \frac{\cancel{\frac{p}{1-x^p}} \cdot (1-x)}{\cancel{\frac{q}{1-x^q}} \cdot (1-x)} -$$

$$= \lim_{x \rightarrow 1} \left(\frac{p(1-x^q) - q(1-x^p)}{(1-x^p)(1-x^q)} \right)$$

$$\boxed{\frac{1}{1-x}} \rightarrow \frac{1}{1-x} = 0$$

$$= \lim_{x \rightarrow 1} \frac{p - q - px^q + qx^p}{(1-x^p)(1-x^q)(1-x)^2}$$

$$x = 1 + t$$

$$= \lim_{t \rightarrow 0} \frac{p - q - p(1+t)^q + q(1+t)^p}{p \cdot q \cdot t^2}$$

Ques $\lim_{x \rightarrow \infty} (x - \ln(\cosh(x)))$ where $\cosh(t) = \frac{e^t + e^{-t}}{2}$.

$$= \frac{\left(-\frac{p \cdot q \cdot (q-1)}{2} + \frac{p \cdot q \cdot (p-1)}{2} \right) t^2}{(p \cdot q) t^2} = \boxed{\frac{p-q}{2}} \text{ Ans}$$

$$= \frac{p - q - p \left(1 + qt + \frac{q(q-1)}{2} t^2 \right) + q \left(1 + pt + \frac{p(p-1)}{2} t^2 \right)}{p \cdot q \cdot t^2}$$

$$\lim_{x \rightarrow \infty} (x - \ln(\cosh x))$$

$$\cosh(t) = \frac{e^t + e^{-t}}{2}$$

$$\ln(t)$$

$$= \lim_{x \rightarrow \infty} \left(x - \ln\left(\frac{e^x + e^{-x}}{2}\right) \right)$$

$$= \lim_{x \rightarrow \infty} \left(\ln(e^x) - \ln\left(\frac{e^x + e^{-x}}{2}\right) \right)$$

$$= \lim_{x \rightarrow \infty} \left(\ln\left(\frac{e^x}{\frac{e^x + e^{-x}}{2}}\right) \right) = \lim_{x \rightarrow \infty} \ln\left(\frac{2}{1 + e^{-2x}}\right)$$

$$= \boxed{\ln(2)}$$

$$\boxed{\text{Que}} \lim_{x \rightarrow 0^+} x^x$$

$$= \lim_{x \rightarrow 0^+} e^{x \cdot \ln(x)}$$

$$= \lim_{x \rightarrow 0^+} e^{\frac{\ln(x)}{(1/x)}} \left(\frac{\infty}{\infty} \right)$$

$$= \lim_{x \rightarrow 0^+} e^{\frac{1/x}{-1/x^2}} = e^{-x}$$

$$= e^0$$

$$= \boxed{1}$$

$$\boxed{\text{Que}} \lim_{x \rightarrow 0^+} x^{x^x - 1}$$

$$\text{Soln}$$

$$\lim_{x \rightarrow 0^+} x^{(x^x - 1)}$$

$$= \lim_{x \rightarrow 0^+} e^{(x^x - 1) \cdot \ln(x)}$$

$$= \lim_{x \rightarrow 0^+} e^{\underbrace{(x^x - 1)}_{\rightarrow 0} \ln(x)}$$

$$= \lim_{x \rightarrow 0^+} \exp\left(\frac{e^{x \ln(x)} - 1}{x} \ln(x)\right)$$

$$= \lim_{x \rightarrow 0^+} \exp\left(x \ln(x) \cdot \ln(x)\right)$$

$$= \lim_{x \rightarrow 0^+} \exp\left(x \ln^2(x)\right) = \exp(0)$$

$$= \boxed{1}$$

$$\boxed{\text{Que}} \lim_{x \rightarrow 0^+} x^{x^{x^x}}$$

$$= \lim_{x \rightarrow 0^+} (x)^{x^x}$$

$$= \exp\left(x^x \cdot \ln(x)\right)$$

$$= \exp\left(\frac{x^x}{x} \cdot x \cdot \ln(x)\right)$$

$$= \exp\left(\underbrace{x^{x-1}}_{\rightarrow 1} \cdot \underbrace{x \ln(x)}_{\rightarrow 0}\right)$$

$$= \exp(0) = 1$$

$$\boxed{1}$$

Results to Remember

$$\textcircled{1} \quad \lim_{x \rightarrow 0^+} x^m \cdot \ln(x) = 0, \quad m \in \mathbb{R}^+$$

$\begin{array}{ccc} & \nearrow 0 & \searrow -\infty \\ & \ln(x) & \end{array}$

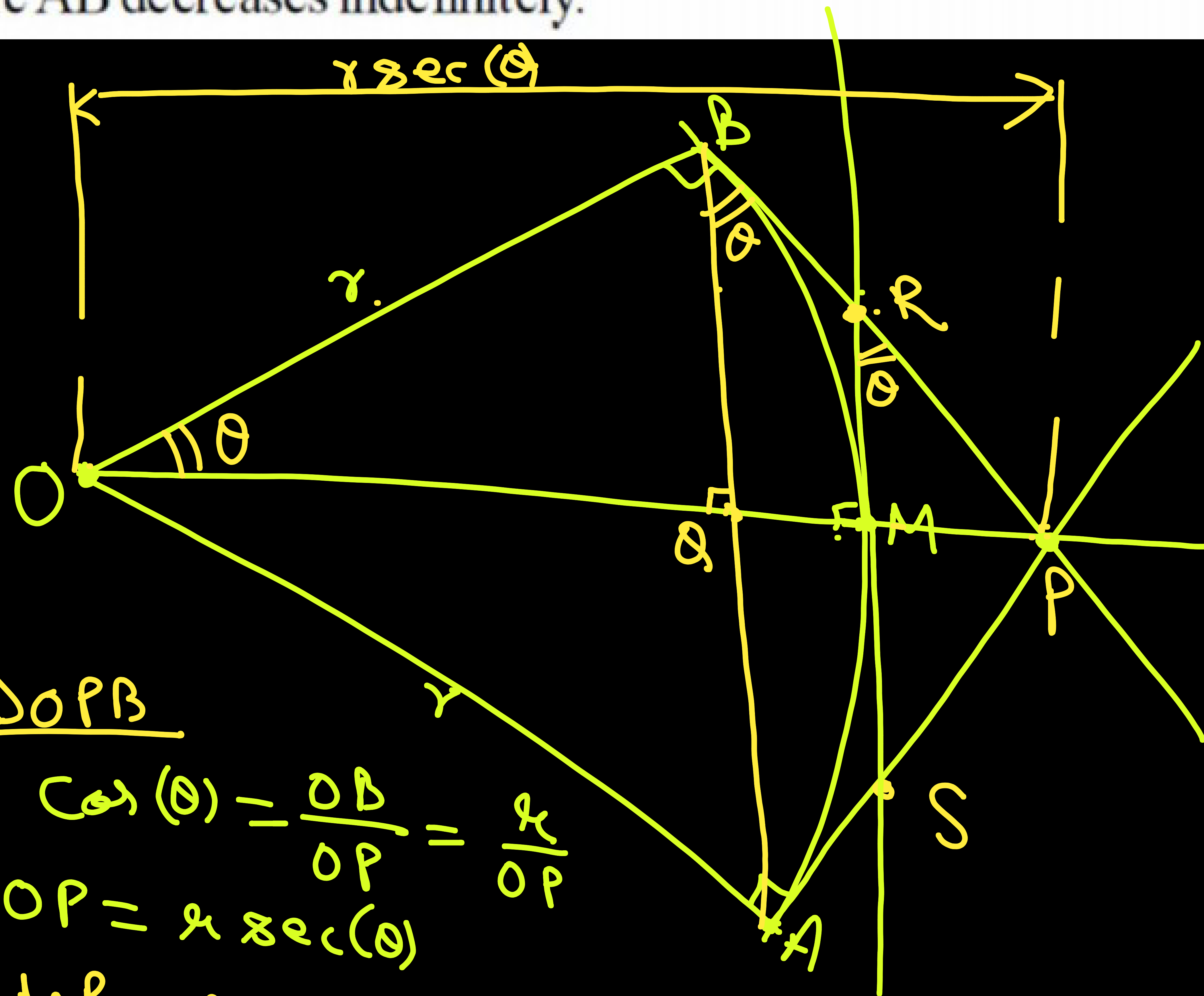
$$\textcircled{2} \quad \lim_{x \rightarrow 0^+} x^x = e^{\frac{x \ln(x)}{1}} = e^0 = 1.$$

$$e^{A \rightarrow 0^+} - 1 \longrightarrow A$$

$$\begin{aligned} a^m &= e^{\ln(a^m)} \\ &= e^{m \ln(a)} \\ &= \exp(m \ln(a)) \end{aligned}$$

Ques

At the end-points and ^{at} the midpoint of a circular arc AB tangent lines are drawn, and the points A and B are joined with a chord. Prove that the ratio of the areas of the two triangles thus formed tends to 4 as the arc AB decreases indefinitely.



In $\triangle OPB$

$$\cos(\theta) = \frac{OB}{OP} = \frac{r}{OP}$$

$$OP = r \sec(\theta)$$

$$MP = r \sec(\theta) - r$$

In $\triangle MPR$ $\tan(\theta) = \frac{MP}{MR} \Rightarrow MR = MP \cot(\theta)$

In $\triangle OBQ$ $BQ = r \sin(\theta)$

In $\triangle PQB$ $\tan(\theta) = \frac{PQ}{BQ} \Rightarrow PQ = BQ \tan(\theta)$

$$\frac{Ar(\triangle PAB)}{Ar(\triangle PRS)} = \frac{\frac{1}{2} \cdot PQ \cdot 2BQ}{\frac{1}{2} \cdot PM \cdot 2RM} = \frac{BQ \tan(\theta) \cdot BQ}{MP \cdot MP \cot(\theta)}$$

$$= \frac{BQ^2 \cdot \tan^2(\theta)}{MP^2}$$

$$= \frac{r^2 \sin^2(\theta) \cdot \tan^2(\theta)}{r^2 (\sec(\theta) - 1)^2} = \frac{\sin^4(\theta) \cancel{\cos^2(\theta)}}{\cancel{\cos^2(\theta)} \cdot (1 - \cos(\theta))^2}$$

Now $\widehat{AB} \rightarrow 0 \Rightarrow 0 \rightarrow 0.$

$$\lim_{\theta \rightarrow 0} \frac{A_r(\Delta PAB)}{A_r(\Delta PRS)} = \lim_{\theta \rightarrow 0} \frac{\sin^4(\theta)}{(1 - \cos(\theta))^2}$$

$$= \lim_{\theta \rightarrow 0} \frac{\theta^4}{\left(\frac{\theta^2}{2}\right)^2} = \lim_{\theta \rightarrow 0} \frac{4 \cdot \cancel{\theta^2}}{\cancel{\theta^2}}$$

$$= \boxed{4}$$

$$1 - \cos(A) \rightarrow A^2/2$$

Ques Let $P_n = a^{P_{n-1}} - 1$, $\forall n \geq 2$ and $P_1 = a^x - 1$ then evaluate

$$\lim_{x \rightarrow 0} \frac{P_{2021}}{x}$$

Sol $P_1 = \frac{a^x - 1}{P_{n-1} - 1}$

$$P_n = a^{P_{n-1}} - 1$$

$$P_2 = a^{P_1} - 1$$

$$P_3 = a^{P_2} - 1$$

$$x \rightarrow 0 \Rightarrow P_1 \rightarrow 0 \Rightarrow P_2 \rightarrow 0 \Rightarrow P_3 \rightarrow 0$$

$$\lim_{x \rightarrow 0} \frac{P_{2021}}{x}$$

$$= \lim_{x \rightarrow 0} \frac{P_{2021}}{P_{2020}} \cdot \frac{P_{2020}}{x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{a^{P_{2020}} - 1}{P_{2020}} \right) \cdot \lim_{x \rightarrow 0} \frac{P_{2020}}{x}$$

$$= \ln(a) \cdot \lim_{x \rightarrow 0} \frac{P_{2020}}{x} = \ln^2(a) \cdot \lim_{x \rightarrow 0} \frac{P_{2019}}{x}$$

$$= \ln^{2020}(a) \lim_{x \rightarrow 0} \frac{P_1}{x} = \ln^{2020}(a) \lim_{x \rightarrow 0} \frac{a^x - 1}{x}$$

$$\lim_{t \rightarrow 0} \frac{a^t - 1}{t} = \ln(a)$$

$$\ln^{2021}(a)$$

Ans

Que

$$\lim_{n \rightarrow \infty} \left(\frac{\sqrt{n^2+n} - 1}{n} \right)^{2\sqrt{n^2+n} - 1}$$

Sol

$$\lim_{n \rightarrow \infty} \left(\frac{\sqrt{n^2+n} - 1}{n} \right)^{2\sqrt{n^2+n} - 1} = \exp \left(\lim_{n \rightarrow \infty} (2\sqrt{n^2+n} - 1) \left(\frac{\sqrt{n^2+n} - 1}{n} - 1 \right) \right)$$

$$= \exp \left(\lim_{n \rightarrow \infty} (2\sqrt{n^2+n} - 1) \left(\frac{\sqrt{n^2+n} - (1+n)}{n} \right) \right)$$

$$= \exp \left(\lim_{n \rightarrow \infty} \frac{(2n)}{n} \left(\frac{n^2+n - (1+n)^2}{\sqrt{n^2+n} + 1+n} \right) \right) = \exp \left(\lim_{n \rightarrow \infty} \frac{n^2+n-1-n^2-2n}{2n} \right)$$

$$\lim_{x \rightarrow a} (f)^g \rightarrow \exp \left(\frac{(f-1) \cdot g}{1} \right)$$

$$= \exp(-1) = \boxed{e^{-1}}$$

Ans

Ques

$$\begin{aligned} & \lim_{n \rightarrow \infty} 2^{n+1} \cdot \sqrt{2 - x_n} \\ &= \lim_{n \rightarrow \infty} 2^{n+1} \cdot \sqrt{2 - 2 \cos\left(\frac{\theta}{2^n}\right)} \\ &= \lim_{n \rightarrow \infty} 2^{n+1} \sqrt{4 \sin^2\left(\frac{\theta}{2^{n+1}}\right)} \\ &= \lim_{n \rightarrow \infty} 2^{n+1} \cdot 2 \cdot \sin\left(\frac{\theta}{2^{n+1}}\right) \\ &= \lim_{n \rightarrow \infty} 2^{n+1} \cdot 2 \cdot \frac{\theta}{2^{n+1}} = 2\theta = 2 \cdot \frac{\pi}{6} = \boxed{\frac{\pi}{3}} \end{aligned}$$

Ques

Let $x_0 = 2 \cos \frac{\pi}{6}$ and $x_n = \sqrt{2 + x_{n-1}}$, $n = 1, 2, 3, \dots$. Find $\lim_{n \rightarrow \infty} 2^{n+1} \sqrt{2 - x_n}$.

$$\text{Let } x_0 = 2 \cos\left(\frac{\pi}{6}\right) = 2 \cos(0).$$

$$\text{Q } x_n = \sqrt{2 + x_{n-1}}, \quad n = 1, 2, 3, \dots$$

$$\text{find } \lim_{n \rightarrow \infty} 2^{n+1} \sqrt{2 - x_n}.$$

Sol

$$x_n = \sqrt{2 + x_{n-1}} \quad \left\{ \begin{array}{l} x_0 = 2 \cos(0) \\ x_1 = \sqrt{2 + x_0} = \sqrt{2 + 2 \cos(0)} = 2 \cos\left(\frac{0}{2}\right) \\ x_2 = \sqrt{2 + x_1} = \sqrt{2 + 2 \cos\left(\frac{0}{2}\right)} = 2 \cos\left(\frac{0}{2^2}\right) \\ \vdots \\ x_n = 2 \cos\left(\frac{0}{2^n}\right) \end{array} \right.$$